ML - Modeling

Monday, March 09, 2026 12:39 PM

Ml : takes data as input : perform analysis and return the predictions
Supervised: we give the input and predict depended variable and target variable e.g. : based on data we do analysis on diabetic
Unsupervised: we don't give label for data we expect the model to determine the output eg : we make the clustering
Semi supervised(or hybrid): half label and half we don't data to the model
Reinforced model: give positive and negative like feedback like the positive and negative eg : self-driving cars

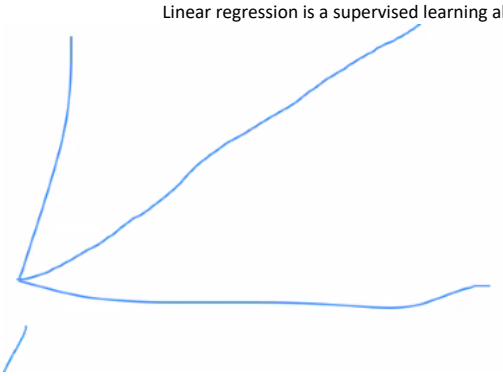
Supervised :**Supervised Learning** is a type of **machine learning where the model learns from labeled data. EG:**The computer learns from **examples where the correct answer is already known.**

Regression: predicts a **continuous numerical value**.

Classification: predicts a **category or class label**. Classify Yes or No , 0 or 1

Regression:

-Linear regression : It finds the best straight line that shows the relationship between input and output.
Linear regression is a supervised learning algorithm used to predict a continuous



Y = mx + c (linear curve)

value by modeling the linear relationship between input variables and output.

Y = mx + c

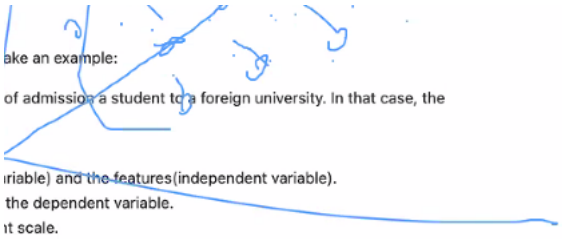
Life cycle of building a ML Model:(Flow is same for linear and Logistic regression

- Data Selections
- Perform EDA(data graph)
- Selection of ML algorithm
- Train and test data
- Validation (seen data vs unseen data
- Tuning hyperparameter of model
- Final model – save the model with .pckl file and deploy
- Perform inferencing
- UI
- consume

After the model is trained and tested, we **give it new input data and it predicts the output.** This process is called **inference**.

/ar able) based on the features(Independent Variables) at hand. Regression is used
n the variables and forecasting. For example, the relationship between the stock
mpany annual performance etc. can be studied using regression.

Here, we fit a curve/line to the data points, in such a manner that the differences
is minimum. The topic will be explained in detail in the coming sections.



Residual : we try to reduce the residual distance during linear regression
(I.e.) : Loss Function – goal of linear reg is to reduce this loss

How to reduce the loss: Gradient decent
Local minima
Local maxima
Global minima
Global maxima

When the model sees all the training data once, that is one epoch.

Example

Suppose you have **1000 training samples**.

- After the model processes all 1000 samples **once → 1 epoch**
- If it processes them **10 times → 10 epochs**

Why we use multiple epochs

The model learns gradually.
Each epoch helps the model **adjust its weights and improve predictions**.

Example:

- Epoch 1 → model learns basic pattern
- Epoch 5 → better accuracy
- Epoch 20 → very accurate model

Epoch is one full pass of the dataset, batch is a subset of data, and iteration is one model update using one batch.

Now we have the best fit line

R^2 statistics: **It tells us how much of the variation in the data is explained by the model.**

how close the predicted values are to the actual values.

- **Higher R^2 → better fit**
- **Lower R^2 → poor fit**

More variables in eqution : the R^2 increases too so more independent values doesnt mean better fit

To handle this : Adjusted R^2 – penalize R^2

Regularizations also handle feature selection

- Laso
- Ridge

X-Hyper parameter tuning – tuning the residual data-X